

Exp 1

Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

code:

```
data=[
    ['High', 'High', 'High', 'Placed'],
    ['High', 'High', 'Medium', 'Placed'],
    ['High', 'Medium', 'Medium', 'Placed'],
    ['Medium', 'Medium', 'Medium', 'Not Placed'],
    ['Low', 'Low', 'Low', 'Not Placed'],
    ['High', 'High', 'High', 'Placed'],
    ['Medium', 'Medium', 'Medium', 'Not Placed'],
    ['High', 'High', 'Medium', 'Placed'],
    ['Low', 'Low', 'Low', 'Not Placed'],
    ['High', 'High', 'High', 'Placed']
]
```

```
hypothesis = ['0', '0', '0']
```

```
print("Initial Hypothesis:", hypothesis)
```

```
for sample in data:
    if sample[-1] == "Placed":
        for i in range(len(hypothesis)):
            if hypothesis[i] == '0':
                hypothesis[i] = sample[i]
            elif hypothesis[i] != sample[i]:
                hypothesis[i] = '?'
```

```
print("\nFinal Hypothesis:")
```

```
print(hypothesis)
```

Output:

```
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/yuvas/Downloads/win-g2010-1_3-n_mcd/win/LIB/machine learning lab/exp 1.py
Initial Hypothesis: ['0', '0', '0']

Final Hypothesis:
['High', '?', '?']
>>>
```